INCH-POUND

MIL-R-48664B (AR)
18 March 1993
SUPERSEDING
MIL-L-48664A (AR)
10 August 1988

MILITARY SPECIFICATION

RESTRAINT SYSTEMS, AMMUNITION (USING BLOW MOLDED PLASTIC DIVIDERS)

This specification is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the material, requirements, examinations, tests and packaging for a family of blow molded plastic divider racks used for restraining loose: unpalletized 155mm projectiles; unpalletized 8 inch projectiles; or unpalletized, unboxed 120mm mortar rounds contained in fiber tubes during on-vehicle storage and transport (see 6.1). The rack consists of specifically designed plastic dividers that, when assembled, will restrain any conventional projectiles of the 155mm or 8 inch family, or all 120mm mortar rounds in fiber tubes (see 6.4). The loaded racks are restrained in a tactical transport vehicle by means of a cargo tiedown strap(s) applied in an approved manner. The combinations for 155mm or 8 inch are referred to as Loose Projectile Restraint System (LPRS). The combination for 120mm mortar is referred to as Loose Round Restraint System (LRRS).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 2590

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

MIL-A-70625 - Automated Acceptance Inspection Equipment for Non-Electronic Components, Design, Testing and Approval of

STANDARDS

FEDERAL

FED-STD-595 - Color

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions

MIL-STD-129 - Marking for Shipment and Storage

HANDBOOKS

MILITARY

MIL-HDBK-728/6 - Military Handbook, Ultrasonic Testing

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from DODSSP-Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in cited in the solicitation.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:57Z
Check the source to verify that this is the current version before use.

DRAWINGS

ARMY MATERIEL COMMAND

9207909	- Projectile 8-inch, M106HE, LAP Assy
9210140	- Projectile 8-inch, M509A1 ICM, LAP Assy
9215220	- Projectile 155mm, M483A1 DPICM, LAP Assy
9216352	- Projectile 155mm, M107 HE, LAP Assy
9235999	- Projectile 155mm, M549A1 HE RAP, LAP Assy
9287993	- Projectile, 8-inch, M650 RAP, LAP Assy
12577501	- Cartridge 120mm, HE, M934
12577502	- Cartridge 120mm, Smoke, M929
12577503	- Cartridge 120mm, Illum., M930
12577504	- Cartridge 120mm, HE, M933
AC200000400	- Combination of Adapted Items, Loose Projectile Restraint System, 155mm
AC200000401	- Rack, Ammunition, 15 Round Assy, Loose Projectile Restraint System, 155mm
AC200000402	- Divider, Short, Loose Projectile Restraint System, 155mm
AC200000403	- Divider, Long, Loose Projectile Restraint System, 155mm
AC200000404	- Packaging Drawing, Loose Projectile Restraint System, 155mm

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

AC200000405 - Test, Ease of Disassembly, Loose Projectile Restraint System, 155mm

AC200000408 - Material for Divider, Ammunition, Loose Projectile Restraint System

AC200000438 - Divider, Ammunition, Loose Projectile Restraint System, 8-inch (203mm)

AC200000571 - Combination of Adapted Items, Loose Projectile Restraint System, 8-in (203mm)

AC200000572 - Rack, Ammunition, 9 Round Assy, Loose Projectile Restraint System, 8-in (203mm)

AC200000573 - Packaging Drawing, Loose Projectile Restraint System, 8-inch (203mm)

AC200000574 - Test, Ease of Disassembly, Loose Projectile Restraint System, 8-inch (203mm)

AC200000591 - Divider, Ammunition, Loose Round Restraint System (LRRS), 120mm Mortar

AC200000596 - Combination of Adapted Items, Loose Round Restraint System (LRRS), 120mm Mortar Rds in Fiber Tubes

AC200000597 - Rack, Ammunition, 36 Round Assy, LRRS, 120mm Mortar Rds in Fiber Tubes

AC200000598 - Packaging Drawing, Rack, Ammunition, 36 Rd Assy, LRRS, 120mm Mortar

AC200000599 - Test, Ease of Disassembly, Loose Round Restraint System (LRRS) 120mm Mortar

TECHNICAL MANUALS

MILITARY

TM 9-2590-210-10 - Operators Manual: Loose Projectile Restraint System (LPRS) for use with Field Artillery Companion Vehicles

(Copies of other Government documents, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. The materials used in the manufacture of the applicable divider(s), ammunition, shall be as specified on Dwg AC200000408. Certification of compliance shall be provided by the contractor.

3.2 Dividers. Parts and assemblies shall comply with all requirements specified on the applicable drawings and with all requirements specified in the applicable specifications.

3.2.1 Flatness. When new and in a relaxed position laid flat on a side, the sides of a divider shall be straight within 1/8 inch from a 24 inch straight edge placed at any location other than over the stacking lugs.

3.2.2 Weld cones. Weld cones shall be homogeneously welded together at the plane of the mold parting line to provide maximum strength to the divider as intended by design.

3.2.3 Drainage holes. Drainage will be provided for any moisture accumulated within the divider interior when the dividers are assembled on either edge.

3.2.4 Color. The color of the divider shall be olive drab matching any of the following colors of Federal Standard 595: -34052, -34079, -34086, -34087, -34096, or -34102. To retard sun degradation, ultraviolet stabilizer shall be added into the color concentrate. Certification shall be provided by the contractor to assure that the ultraviolet stabilizer added into the color concentrate consists of 0.3 percent Tinuvin 770 and 0.05 percent Igranox 1076 based on a 4.0 percent color letdown.

3.2.5 Stacking lugs. Dividers when laid flat and stacked, will nest on their stacking lugs without space between dividers. This nesting is required in any flat position, i.e., open slots pointing in the same direction or alternating directions, NSN up or down. The vertical sides of the stack shall align within + 1/8 inch. Different lengths of dividers are not intermixed for this requirement (see 4.5.6).

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:57Z
Check the source to verify that this is the currentversion before use.

3.3 Performance test. The following defects shall not be permitted on the tested assemblies:

a. Failure of the rack by deforming, splitting or breaking, which allows the dividers to disengage and spill the projectiles/rounds during the performance tests.

b. Separating of the parting line seam more than 2.5 inches along the top edge of the divider in contact with the projectiles/rounds. This failure shall not result in a collapse of the divider walls below the next structural stiffener; either handle, hand grip or weld cone.

c. Any breaking or cracking of the rack other than at the parting line seam (see 3.3.b.) during the performance tests. Punctures, abrasions and wear though caused by the restraint strap metal parts or projectile grommets are not counted as a failure, providing such wear does not lead to functional failure of the dividers or rack assembly.

d. Excessive bending of the short sections at the corners of an assembled rack loaded with projectiles/rounds and restrained to a vehicle side wall by a cargo strap on three sides of the rack per TM 9-2590-210-10 or other applicable restraint publication. Excessive bending is deformation which allows the restraint strap(s) to disengage from the strap slot allowing the rack assembly to radically shift or upset during the performance tests (see 4.5.1.2.).

e. Deformation of a divider which interferes with ease of disassembly (see 4.5.5).

f. Failure of disassembled dividers to be reassembled into a load bearing rack for successful continuation of the performance test.

3.4 Divider markings. The divider design shall include a thirteen digit (plus three dashes) National Stock Number (NSN) molded in a recessed manner in each divider in 3/8 inch characters by 1/16 inch deep on one side only. The divider design shall include a material designator code and year (last 2 digits) the divider was manufactured molded in a recessed manner in each divider in 1/4 characters by 1/32 deep on one side only. The divider design shall include the contract number molded in a recessed manner in

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

each divider in 1/4 characters by 1/32 deep on the opposite side from and same location as the NSN. See drawing AC200000408 for material designator. See applicable part drawing for location of markings.

3.5 Ease of assembly and disassembly. The divider slots shall be so constructed to permit ease of assembly and disassembly yet the assembled divider rack must be rigid enough to maintain its original configuration while loaded. The loaded rack configuration is evaluated by a performance test (see 4.5.1) and extreme operating temperature tests (see 4.5.4). Ease of disassembly is tested as specified in 4.5.5.

3.6 Extreme operating temperatures. The rack, ammunition, must restrain the applicable ammunition load at temperatures from minus 60 degrees Fahrenheit to plus 145 degrees Fahrenheit (see 4.5.4), as evidenced by meeting the performance test requirements of 3.3.

3.7 Process controls. The contractor's quality program plan or detailed inspection plan shall include documented process controls which detail the specific procedures involved in the manufacture of the plastic dividers. These documented process controls shall be submitted to the contracting officer for review and approval by the cognizant technical agency (CTA) in accordance with the approved DID specified on the Contract Data Requirements List (CRDL) in the contract (see 6.3). The contractor shall not initiate production without written approval from the contracting officer. Once approved, the contractor shall not change the process without written approval from the contracting officer. The documented process controls shall include the following information as a minimum:

a. Item nomenclature

b. Applicable component (divider)

c. List of applicable drawings and specifications

d. List of inspection equipment

e. List of materials and suppliers

f. Process flow charts

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

g. Methods for controlling (and documenting) operational parameters of process (examples; temperature, pressure, feed times, rate of mold closure).

h. Procedure for mixing virgin resin material and reground material and method for identifying these batches or continuous flow.

3.8 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.9 Workmanship. The dividers shall be regular, relatively smooth and free from wrinkles and rough spots to the extent that minor irregularities shall not interfere with successful testing under 4.5.5. Critical areas are within and adjacent to the assembly slots. The dividers shall be free from cracks, pinholes and any other defects that might affect serviceability and durability. Flashing at the parting lines shall not exceed 1/16 inch height on external perimeters and 1/32 inch height at the handles (if applicable). Flashing in the open slots shall be removed uniformly to meet tolerances of the open slot. Clean up shall include breaking of sharp corners. All dividers shall be free of chips, dirt, grease and foreign material. If cleaning agents are used, the method shall not be injurious to the divider material nor shall any of the dividers be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-A-48078 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.1.3 Certification.

4.1.3.1 Certification of material (see 3.1). A certification of material conformance shall be submitted by the contractor for each lot utilized, confirming the two basic material parameters which dictate the physical and chemical characteristics of the polyolefin materials as listed on Dwg AC200000408. Specifically, the two basic material parameters are: melt flow and density. Additionally, the latest nominal physical properties for the parameters listed are to be supplied with the certification. The basic and audit data shall be found to be compatible with the parameters as listed on Dwg AC200000408. The method of certification shall contain as a minimum the information as stated on Dwg AC200000408. The submission of the documentation shall be as specified in the contract.

4.1.3.2 Certification of ultraviolet stabilizer. A certificate of material conformance shall be submitted by the contractor, certifying that the ultraviolet stabilizer, added to the color concentrate, meets the requirements of 3.2.4. The method of certification shall contain as a minimum the information as stated on Dwg AC200000408. The submission of the documentation shall be as specified in the contract.

4.2 Classification of inspections. The following types of inspection shall be conducted on these items:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of twenty (20) sets of divider racks or as specified in the contract (see 4.3.1.1) which shall have been produced by the contractor using the same production processes,procedures, dimensional tolerances and equipment as will be used in fulfilling the contract. All materials shall be obtained from the same source of supply as will be used in regular production.

4.3.1.1 Number of dividers in sets. When the contract specifies submission by sets, the number of individual dividers per set is as follows:

a. 155mm LPRS: 6 divider, ammunition, short, LPRS
4 divider, ammunition, long, LPRS

b. 8-inch LPRS: 8 divider, ammunition, LPRS

c. 120mm mortar: 14 divider, ammunition, LRRS

4.3.1.2 Percent regrind for first article submission. To insure that production processes, procedures, and material are the same for First Article Submission as in production, the contractor shall first determine the percent regrind which will be used in production. This percentage shall not exceed 45 percent. Prior to saving any sample for First Article Submission, the material mix to the extruder will be run through 5 iterations to insure the first sample saved contains less than 2 percent of material that has seen five or more melt cycles. For purposes of establishing the regrind mix through five iterations, flashing and parts may be used for regrind material. See 6.4.5 for a discussion of the method for obtaining the fifth iteration at 45 percent regrind material.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I.

4.3.3 Rejection. See MIL-A-48078.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-R-48664B (AR)

PARAGRAPH 4.3.3.1	TITLE Divider, Ammunition, Short, 155mm LPRS Divider, Ammunition, Long, 155mm LPRS Rack, Ammunition, 15 Round Assy, 155mm LPRS	SHEET 1 OF 3	DRAWING NUMBER See Below NEXT HIGHER ASSEMBLY AC200000400

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Divider, Ammunition, Short See Applicable Drawing: AC200000402			
	a. Examination for compliance	28	3.2	4.4.2.1
	b. Destructive test for conformation of weld cone weldment	4	3.2.2	4.5.3
	Divider, Ammunition, Long See Applicable Drawing: AC200000403			
	a. Examination for compliance	20	3.2	4.4.2.2
	b. Destructive test for conformation of weld cone weldment	4	3.2.2	4.5.3
	Rack, Ammunition, 15 Round Assembly See Applicable Drawing AC200000401/405			
	a. Ease of assembly and disassembly	6 Racks	3.5	4.5.5
	b. Performance test			
	(1) 9 round rack configuration	2 Racks		
	(2) 15 round rack configuration	2 Racks		
	(3) 25 round rack configuration	1 Rack		
	c. Extreme operating temperature test (15 round rack assembly only)	4 Racks	3.6	4.5.4
	(extra dividers, short)	6		
	(extra dividers, long)	6		

NOTES: Conditioning: The dividers shall be conditioned at ambient temperature for a minimum of 24 hours before any inspection of drawing dimensions is performed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-R-48664B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.3.2	Divider, Ammunition, 8-inch LPRS Rack, Ammunition, 9 Round Assy, 8-inch LPRS	2 OF 3	See Below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Divider, Ammunition, See Applicable Drawing: AC200000438			
	a. Examination for compliance	28	3.2	4.4.2.4
	b. Destructive test for conformation of weld cone weldment	4	3.2.2	4.5.3
	Rack, Ammunition, 9 Round Assembly See Applicable Drawing AC200000572/574			
	a. Ease of assembly and disassembly	6 Racks	3.5	4.5.5
	b. Performance test	2 Racks	3.3	4.5.1
	c. Extreme operating temperature test (extra dividers)	4 Racks 6	3.6	4.5.4

NOTES: Conditioning: The dividers shall be conditioned at ambient temperature for a minimum of 24 hours before any inspection of drawing dimensions is performed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-R-48664B (AR)

PARAGRAPH 4.3.3.3	TITLE Divider, Ammunition, 120mm Mortar, LRRS; Rack, Ammunition, 36 Round Assy, 120mm Mortar, LRRS	SHEET 3 OF 3	DRAWING NUMBER See Below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Divider, Ammunition, 120mm Mortar See Applicable Drawing: AC200000591			
	a. Examination for compliance	28	3.2	4.4.2.6
	b. Destructive test for conformation of weld cone weldment	4	3.2.2	4.5.3
	Rack, Ammunition, 9 Round Assembly See Applicable Drawing AC200000597/599			
	a. Ease of assembly and disassembly	6 Racks	3.5	4.5.5
	b. Performance test	2 Racks	3.3	4.5.1
	36 round rack configuration			
	c. Extreme operating temperature test	4 Racks	3.6	4.5.4
	(extra dividers)	6		

NOTES: Conditioning: The dividers shall be conditioned at ambient temperature for a minimum of 24 hours before any inspection of drawing dimensions is performed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All materials submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with the requirements.

4.4.2 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table II below, using the inspection levels stated in the Classification of Characteristics paragraphs.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

TABLE II. Attributes sampling inspection

Lot Size	Inspection Levels III	IV
2 to 8	*	*
9 to 15	*	13
16 to 25	*	13
26 to 50	32	13
51 to 90	32	13
91 to 150	32	13
151 to 280	32	32
281 to 500	32	32
501 to 1200	80	50
1201 to 3200	80	50
3201 to 10000	125	50
10001 to 35000	125	80
35001 to 150000	125	80
150001 to 500000	200	125
500001 to over	200	125

Numbers under inspection levels indicate sample size; asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.9). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provision(s) herein. In cases of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

MIL-R-48664B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Divider, Ammunition, Short, 155mm LPRS	1 OF 1	AC200000402
			NEXT HIGHER ASSEMBLY: AC200000401

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Fold lines	Level III	3.2	4.5.7/Visual
102	Small width of open slot (0.625)	Level III	3.2	SMTE (see 6.4)
103	Large width of open slot (1.190)	Level III	3.2	SMTE
104	Thickness of open slot (0.575)	Level III	3.2	SMTE
105	Width of closed slot (0.625)	Level III	3.2	SMTE
106	Thickness of closed slot (0.575)	Level III	3.2	SMTE
107	Thickness of divider both sides of slot, closed edge to center line (1.130).	Level III	3.2	SMTE
108	Distance center to center of slot (8-1/8)	Level III	3.2	SMTE
109	Location and dimensions of stacking lugs.	Level III	3.2.5	4.5.6/SMTE
110	Flatness	Level III	3.2.1	4.5.6/SMTE
111	Wall thickness	Level III	3.2	4.5.2/SMTE
Minor				
201	Overall length of divider	Level IV	3.2	SMTE
202	Overall height of divider	Level IV	3.2	SMTE
203	Color incorrect	Level IV	3.2.4	Visual
204	Drainage holes missing	Level IV	3.2.3	Visual
205	Legibility of markings	Level IV	3.4	Visual
206	Workmanship	Level IV	3.8	Visual/SMTE

NOTES: Conditioning: The dividers shall be conditioned at ambient temperature a minimum of 24 hours before any inspection of drawing dimensions is performed.

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T00:57Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-R-48664B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Divider, Ammunition, Long, 155mm LPRS	1 OF 1	AC200000403
			NEXT HIGHER ASSEMBLY: AC200000401

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Fold lines	Level III	3.2	4.5.7/Visual
102	Small width of open slot (0.625)	Level III	3.2	SMTE (see 6.4)
103	Large width of open slot (1.190)	Level III	3.2	SMTE
104	Thickness of open slot (0.575)	Level III	3.2	SMTE
105	Width of closed slot (0.625)	Level III	3.2	SMTE
106	Thickness of closed slot (0.575)	Level III	3.2	SMTE
107	Thickness of divider both sides of slot, closed edge to center line (1.130).	Level III	3.2	SMTE
108	Distance center to center of slot (8-1/8)	Level III	3.2	SMTE
109	Location and dimensions of stacking lugs.	Level III	3.2.5	4.5.6/SMTE
110	Flatness	Level III	3.2.1	4.5.6/SMTE
111	Wall thickness	Level III	3.2	4.5.2/SMTE
Minor				
201	Overall length of divider	Level IV	3.2	SMTE
202	Overall height of divider	Level IV	3.2	SMTE
203	Color incorrect	Level IV	3.2.4	Visual
204	Drainage holes missing	Level IV	3.2.3	Visual
205	Legibility of markings	Level IV	3.4	Visual
206	Workmanship	Level IV	3.8	Visual/SMTE

NOTES: Conditioning: The dividers shall be conditioned at ambient temperature for a minimum of 24 hours before any inspection of drawing dimensions are performed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-R-48664B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Rack, Ammunition, 15 Round Assembly, 155mm, LPRS	1 OF 1	AC200000405
			NEXT HIGHER ASSEMBLY: AC200000401

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Ease of assembly and disassembly	Level III	3.5	4.5.5/SMTE (see 6.5)
Minor	None defined			

NOTES: Conditioning: The dividers shall be conditioned at ambient temperature for a minimum of 24 hours before any inspection of drawing dimensions are performed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:51Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-R-48664B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Divider, Ammunition, 8-inch LPRS	1 OF 1	AC200000438
			NEXT HIGHER ASSEMBLY
			AC200000571

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Fold lines	Level III	3.2	4.5.7/Visual
102	Small width of open slot (0.813)	Level III	3.2	SMTE (see 6.4)
103	Large width of open slot (1.625)	Level III	3.2	SMTE
104	Thickness of open slot (0.750)	Level III	3.2	SMTE
105	Width of closed slot (0.813)	Level III	3.2	SMTE
106	Thickness of closed slot (0.750)	Level III	3.2	SMTE
107	Thickness of divider both sides of slot, closed edge to center line (1.500).	Level III	3.2	SMTE
108	Distance center to center of slot (10-3/4)	Level III	3.2	SMTE
109	Location and dimensions of stacking lugs.	Level III	3.2.5	4.5.6/SMTE
110	Flatness	Level III	3.2.1	4.5.6/SMTE
111	Wall thickness	Level III	3.2	4.5.2/SMTE
Minor				
201	Overall length of divider	Level IV	3.2	SMTE
202	Overall height of divider	Level IV	3.2	SMTE
203	Color incorrect	Level IV	3.2.4	Visual
204	Drainage holes missing	Level IV	3.2.3	Visual
205	Legibility of markings	Level IV	3.4	Visual
206	Workmanship	Level IV	3.8	Visual/SMTE

NOTES: Conditioning: The dividers shall be conditioned at ambient temperature for a minimum of 24 hours before any inspection of drawing dimensions are performed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check this source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-R-48664B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Rack, Ammunition, 9 Round Assembly, 8-inch LPRS	1 OF 1	AC200000574
			NEXT HIGHER ASSEMBLY
			AC200000571

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Ease of assembly and disassembly	Level III	3.5	4.5.5/SMTE (see 6.4)
Minor	None defined			

NOTES: Conditioning: The dividers shall be conditioned at ambient temperature for a minimum of 24 hours before any inspection of drawing dimensions are performed.

Source: https://esdsl.dla.mil Downloaded: 2020-10-20T00:57Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-R-48664B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Divider, Ammunition, 120mm, LRRS	1 OF 1	AC200000591
			NEXT HIGHER ASSEMBLY: AC200000596

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Fold lines	Level III	3.2	4.5.7/Visual
102	Small width of open slot (0.625)	Level III	3.2	SMTE (see 6.4)
103	Large width of open slot (1.190)	Level III	3.2	SMTE
104	Thickness of open slot (0.575)	Level III	3.2	SMTE
105	Width of closed slot (0.625)	Level III	3.2	SMTE
106	Thickness of closed slot (0.575)	Level III	3.2	SMTE
107	Thickness of divider both sides of slot, closed edge to center line (1.130).	Level III	3.2	SMTE
108	Distance center to center of slot (8-1/8)	Level III	3.2	SMTE
109	Location and dimensions of stacking lugs.	Level III	3.2.5	4.5.6/SMTE
110	Flatness	Level III	3.2.1	4.5.6/SMTE
111	Wall thickness	Level III	3.2	4.5.2/SMTE
Minor				
201	Overall length of divider	Level IV	3.2	SMTE
202	Overall height of divider	Level IV	3.2	SMTE
203	Color incorrect	Level IV	3.2.4	Visual
204	Drainage holes missing	Level IV	3.2.3	Visual
205	Legibility of markings	Level IV	3.4	Visual
206	Workmanship	Level IV	3.8	Visual/SMTE

NOTES: Conditioning: The dividers shall be conditioned at ambient temperature for a minimum of 24 hours before any inspection of drawing dimensions are performed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-R-48664B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Rack, Ammunition, 36 Round Assembly 120mm Mortar LRRS (In Fiberboard Tubes)	1 OF 1	AC200000599
			NEXT HIGHER ASSEMBLY AC200000596

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Ease of assembly and disassembly	Level III	3.5	4.5.5/SMTE (see 6.4)
Minor	None defined			

NOTES: Conditioning: The dividers shall be conditioned at ambient temperature for a minimum of 24 hours before any inspection of drawing dimensions are performed.

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

4.4.3 Inspection equipment. The inspection equipment required to perform the inspections specified herein is identified in the "Inspection Method Reference" column of the Classification of Characteristics listings starting with 4.4.2.1. When Automated Acceptance Inspection Equipment is to be used, the provisions of MIL-A-70625 shall apply. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract. Designs which provide variable measurements instead of attributes data are preferred in order to facilitate the use of statistical process control. See MIL-A-48078 and 6.6 herein.

4.5 Methods of inspection.

4.5.1 Performance test (First article only). The test shall be performed at an approved Government facility (see 6.7). A five (5) ton tactical cargo truck shall be used as the test vehicle. The various loads shall be arranged within the assembled rack configurations on the test vehicle and restrained by previously approved tiedown procedures (see 6.8).

4.5.1.1 Test load. The test load will consist of a number of inert loaded projectiles or rounds as designated in 4.5.1.3. See Table III for physical data description of the ammunition model number.

TABLE III. Physical characteristics of ammunition model numbers for testing restraint systems

Model Number	Caliber	Drawing Number	Weight, pounds	Location of CG, from vehicle bed or vibration table, inches
M107	155mm	9216352	95.0	9.45
M483	155MM	9215220	103.3	13.20
M549	155MM	9235999	95.2	12.82
M106	8-inch	9207909	204.0	
M509A1	8-inch	9210140	207.7	
M650	8-inch	9287993	200.0	
M933 (HE)	120mm	12577504	32 approx	_____
M929 (Smoke)	120mm	12577502	32 approx	_____
M930 (Illum)	120mm	12577503	32 approx	_____

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:57Z
Check the source to verify that this is the current version before use.

4.5.1.2 Road hazard course. The test course consists of two groups of alternative railroad ties spaced 8 to 10 feet apart within the group and 75 feet between the two groups. This course induces a 1-1/2 g vertical force and a 2 g lateral force to the test load. The initial tightening of the cargo restraint strap approximates 500 pounds tension and resultant initial force on the corners of the rack assembly.

4.5.1.3 Test Configurations. Use appropriate Table.

TABLE IV. Road hazard course, 155mm projectiles

Test	Rack Configuration	Rack location	Test load each rack	Test dividers 1/ Short	Long
1	2 ea 9 round	Each side of vehicle over wheels	3 ea M483 3 ea M549 3 ea M107		16
2	2 ea 15 round	Each side of vehicle over wheels, long dividers parallel to side of vehicle	6 ea M483 3 ea M549 6 ea M107	12	8
3	1 ea 25 round	Against left side of vehicle over wheels	10 ea M483 5 ea M549 10 ea M107	12	
			Total Samples	28	20

1/ Similar rounds are placed in adjacent locations laterally across the width of the vehicle.

TABLE V. Road hazard course, 8-inch projectile

Test	Rack Configuration	Rack location	Test load each rack	Test dividers 1/
1	2 ea 9 round	Each side of vehicle over wheels	3 ea M106 3 ea M509A1 3 ea M650	16

1/ Similar rounds are placed in adjacent locations laterally across the width of the vehicle.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:57Z
Check the source to verify that this is the current version before use.

TABLE VI. Road hazard course, 120mm mortar rounds in fiber tubes

Test	Rack Configuration	Rack location	Test load each rack	Test dividers 1/
1	1 ea 36 round	Left side of vehicle over wheels	12 ea M933 12 ea M929 12 ea M930	14
2	1 ea 36 round	Right side of vehicle over wheels	12 ea M933 12 ea M929 12 ea M930	14
			Total Samples	28

1/ Similar rounds are placed in adjacent locations laterally across the width of the vehicle.

4.5.1.4 Test. Each test will consist of six passes over the road hazard course. At the end of each pass, the dividers will be visually examined without dismantling the load. See 3.3 for acceptance criteria.

4.5.2 Wall thickness. (see quality control drawing sheet of applicable dwgs AC200000402, AC200000403, AC200000438, AC200000591) Wall thickness shall be measured by ultrasonic test methods (MIL-HDBK-728/6 applies). The preferred method shall be contact testing with a suitable thin liquid film used as a couplant. The calibration reference standard will be a flat material of known thickness and matched acoustic properties. Wall thickness readings shall be taken at locations shown on the applicable quality control drawing. The drawing also states passing criteria.

4.5.3 Weld cone destructive testing (First article only).

4.5.3.1 155mm LPRS. Select 4 long and 4 short dividers which have passed the wall thickness test (see 4.5.2). Note that there are 30 weld cone pairs per long divider: 11 of these pairs are perpendicular to the long side, 15 are parallel, and four are circle cones. There are 18 weld cone pairs per short divider: 5 of these are perpendicular to the long side, 9 are parallel, and 4 are circle cones. Mark the dividers for identification of all parts prior to making the following saw cuts. Make 5 parallel saw cuts per 2 long and 2 short dividers so that the cuts traverse the center line of

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

all parallel cones and cross approximately midpoint of perpendicular cones and circular cones. Make 7 perpendicular cuts on the other 2 long dividers (5 per other 2 short dividers) so that the cuts traverse the center line of all perpendicular weld cones and cross approximately midpoint of parallel cones and circle cones. The results of the above will allow 4 examinations of each weld cone pair (8 if each half is considered). All weld cones shall be found to be homogeneously welded together at the plane of the mold parting line.

4.5.3.2 8-inch LPRS. Select four dividers which have passed the wall thickness test (see 4.5.2). Note that there are 21 weld cone pairs per divider (the hand grips are counted as a weld cone pair for this inspection). Nine of these pairs are perpendicular to the long side, 12 are parallel. Mark the dividers for identification of all parts prior to making the following saw cuts. Make 5 parallel saw cuts per two of the divider so that the cuts traverse the center line of all parallel cones and cross approximately midpoint of perpendicular cones. Make 5 perpendicular cuts on the other two dividers so that the cuts traverse the center line of all perpendicular weld cones and cross approximately midpoint of parallel cones. The results of the above will allow 4 examinations of each weld cone pair (8 if each half is considered). All weld cones shall be found to be homogeneously welded together at the plane of the mold parting line.

4.5.3.3 120mm mortar LRRS. Select 4 dividers which have passed the wall thickness test (see 4.5.2). Note that there are 36 weld cone pairs per divider (the hand grips are counted as a weld cone pair for this inspection). Twelve of these pairs are perpendicular to the long side, 24 are parallel. Mark the dividers for identification of all parts prior to making the following saw cuts. Make 5 parallel saw cuts per two of the divider so that the cuts traverse the center line of all parallel cones and cross approximately midpoint of perpendicular cones. Make 8 perpendicular cuts on the other two dividers so that the cuts traverse the center line of all perpendicular weld cones and cross approximately midpoint of parallel cones. The results of the above will allow 4 examinations of each weld cone pair (8 if each half is considered). All weld cones shall be found to be homogeneously welded together at the plane of the mold parting line.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

4.5.4 Extreme operating temperature test (First article only). Extreme operating temperature tests will be conducted by the U.S. Army at a suitable test facility to be determined. Dividers for four rack sets plus an additional six each individual dividers shall be selected from the first article sample and shipped to the designated test site. For 155mm LPRS only; the additional dividers shall consist of 6 short and 6 long dividers (see Table VII).

4.5.4.1 Cold test. Dividers for two rack sets plus three extra dividers (for 155mm LPRS only; 3 extra short and 3 extra long) plus two each INERT projectile (or mortar round) test loads as applicable are cold soaked to minus 60 degrees Fahrenheit for a minimum of 24 hours. The INERT projectile tests loads for 155mm and 8-inch LPRS shall have the same weight, center of gravity and grommets as live rounds. The INERT 120mm mortar rounds in fiberboard packaging tubes for testing 120mm LRRS shall have the same weight, center of gravity and fiberboard tubes as live rounds. See Table VII for test loads and test samples.

4.5.4.1.1 Cold vibration table test. The test dividers necessary to assemble one rack of the appropriate restraint system (see Table VII), are removed from the presoak cold chamber and assembled as a rack and positioned on a low frequency vibration table used for surface transportability testing. The vibration table is located within a cold chamber and on completion of loading, the rack and test load will be reconditioned to minus 60 degrees Fahrenheit and maintained at this temperature throughout the test cycle. The rack is loaded with the designated cold soaked INERT test load (see Table VII) and secured to the vibration table by a 1-3/4 inch nylon cargo strap in a similar manner to that used in a 5-ton tactical truck in securing the load with a long side of the rack (a rectangular shaped rack, e.g. long side, is applicable to 155mm LPRS only; other racks are square) against a side wall and with the cargo strap on three sides of the rack (see 4.5.1 and 6.6). The divider rack shall be centered between two tiedown anchor points that are separated by the distance specified in Table VII and are 4 inches above the floor. The proper tension on the cargo tiedown strap is achieved by first tightening to remove all slack and the four or five clicks more on the take-up device. The test will be run at 5 hertz for a total of 30 minutes (loose ammunition criteria). The test run includes two orientations of the rack. The first orientation is with the side of the rack against the wall parallel to the motion of the vibration table and is to run for twenty (20) minutes. The second orientation at 90 degrees is

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

with the side of the rack against the wall perpendicular to the motion of the vibration table and is to run for ten (10) minutes. The restraint of the cargo straps in the second orientation is directly from the wall and causes more flexing of the dividers on the opposite side of the rack from the wall. This orientation has been found to be more severe. The test load is to be reconditioned to minus 60 degrees Fahrenheit prior to any resumption of testing. This test is then repeated for the second cold rack (see 3.3 for acceptance criteria).

4.5.4.1.2 Cold bend test. The three extra dividers (in addition to the two rack sets) are selected from the 24-hour cold soaked dividers (see 4.5.4.1). For 155mm LPRS only; the cold bend test is performed on 3 long and 3 short dividers. Each divider is to be bent 45 degrees with the bend occurring at an assembly slot. This is accomplished by standing on one-half of the divider, reaching down and bending one end up. A second observer is to note the angle of bend. The test is to be done as soon as possible, or within the cold chamber, to avoid warming of the plastic. These dividers are not to be used for further testing (see 3.3 for acceptance criteria); see Table VII.

4.5.4.2 Hot tests. The hot tests are the same as the cold tests except that the two racks, the excess dividers and two INERT test loads are heat soaked to plus 145 degrees Fahrenheit for a minimum of 24 hours. The hot tests use the rest of the dividers which were selected at 4.5.4; see Table VII.

4.5.4.2.1 Hot vibration table test. Procedure and test criteria are the same as 4.5.4.1.2 except for the 145 degree temperature; see Table VII.

4.5.4.2.2 Hot bend test. Procedure and test criteria are the same as 4.5.4.1.1 except for the 145 degree temperature; see Table VII.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:57Z
Check the source to verify that this is the current version before use.

TABLE VII. Extreme operating temperature test samples and test loads

Restraint system	Test	Dividers per test	No. of tests	Test load per test,	Distance between tiedown points see 4.5.1.1
155MM LPRS	Cold Vibration	4 long 6 short	2	6 - M107 6 - M483 3 - M549	54 inches
	Cold Bend	3 long 3 short	1	N/A	N/A
	Hot Vibration	4 long 6 short	2	6 - M107 6 - M483 3 - M549	54 inches
	Hot Bend	3 long 3 short	1	N/A	N/A
8-inch (203mm) LPRS	Cold Vibration	8 dividers	2	3 - M106 3 - M509A1 3 - M650	46 inches
	Cold Bend	3 dividers	1	N/A	N/A
	Hot Vibration	8 dividers	2	3 - M106 3 - M509A1 3 - M650	46 inches
	Hot Bend	3 dividers	1	N/A	N/A
120mm Mortar LRRS	Cold Vibration	14 dividers	2	12 - M933 12 - M929 12 - M930	54 inches
	Cold Bend	3 dividers	1	N/A	N/A
	Hot Vibration	14 dividers	2	12 - M933 12 - M929 12 - M930	54 inches
	Hot Bend	3 dividers	1	N/A	N/A

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

4.5.5 Ease of assembly and disassembly. This test shall be performed in accordance with applicable Test, Ease of Disassembly drawing listed at 2.1.2. In conducting this test, each divider comprising the test sample shall be used at least once. Because of variation between test sample and multiples of dividers to assemble a complete rack, some dividers may need to be used twice (for 155mm LPRS only). In those situations where only one divider is in production, dividers from an accepted lot of the alternative length may be held back from delivery and used to complete this test.

4.5.6 Flatness and stacking. The test sample shall be laid on a test surface which is flat within 1/32 inch. The measurements of this flat surface shall be found on the applicable quality control drawing and were selected so that stacking lugs pointing down will overhang the test surface. The sample shall be checked for flatness using a 24 inch long straight edge in compliance with 3.2.1. After checking flatness, dividers will be stacked in groups of four, per 3.2.5. The overall height of the stack not including stacking lugs shall be found on the applicable quality control drawing (see 4.5.2.1).

4.5.7 Fold lines. (see quality control drawing sheet of applicable dwgs AC200000402, AC200000403, AC200000438, AC200000591) Critical areas for fold line inspection are designated on the applicable quality control drawing. The drawing includes acceptance criteria (see 6.2.5)

4.5.8 Qualification of reworked or new mold cavity.

4.5.8.1 Qualification of a reworked mold cavity. To qualify a reworked mold cavity after production has started, at least one sample from the mold cavity shall be fully inspected at those area(s) of the part which were affected by the rework and found to meet the requirements of the basic drawing and specification. If the results of the rework are hidden from view, ie., parting line seam weld closure or weld cone weld-off etc., the part shall be cut through to determine that the mold is producing a product equal to the first article sample as tested. Established in-process quality controls which have insured an acceptable part prior to rework may be used for this inspection but must be backed up by the same examination after a curing time of 24 hours has elapsed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

4.5.8.2 Qualification of a new mold cavity. To qualify a new mold cavity after production has started will require a new First Article Inspection for that mold. The sample shall consist of 72 Divider, Ammunition, produced from the mold in question. All provisions of 4.3 relating to the divider in question apply. The government will arrange for the conducting of the Performance Tests, 4.5.1 and the Extreme Operating Temperature Tests, 4.5.4 utilizing the First Article Samples and the alternate dividers (155mm LPRS only) from an acceptable lot from the current production or from a previous production.

5. PACKAGING

5.1 Packing.

5.1.1 Commercially packed.

5.1.1.1 155mm LPRS. Rack, 15 round assemblies shall be packed as a complete unit per Dwg AC200000404 sheets 1 and 2. TM 9-2590-210-10 to be inserted into each pack are government furnished. Rack assembly packs shall be arranged on a special low profile skid base for shipment as shown on Dwg AC200000404 sheet 3. The low profile skid base is detailed on sheet 4. Individual dividers when ordered as a single item shall be arranged on a pallet as shown on Dwg AC200000404 sheets 5 and 6. Individual long and short dividers shall not be mixed on a pallet.

5.1.1.2 8-inch LPRS. Rack, 9 round assemblies shall be packed as a complete unit per Dwg AC200000573, sheet 1 of 3. Rack assembly packs shall be arranged on a special low profile skid base for shipment as shown on Dwg AC200000573, sheet 2 of 3. The low profile skid base is detailed on sheet 3. Individual dividers when ordered as a single item shall be arranged on a pallet as shown by note on sheet 2.

5.1.1.3 120mm LRRS. Rack, 36 round assemblies shall be packed as a complete unit per Dwg AC200000598, sheet 1 of 3. Rack assembly packs shall be arranged on a special low profile skid base for shipment as shown on Dwg AC200000598, sheet 2 of 3. The low profile skid base is detailed on sheet 3. Individual dividers when ordered as a single item shall be arranged on a pallet as shown by note on sheet 2.

5.2 Marking. Marking shall be in accordance with MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanation nature that may be helpful, but is not mandatory.)

6.1 Intended use.

6.1.1 155mm and 8-inch LPRS. The intent of the Loose Projectile Restraint System (LPRS) is to provide reusable dunnage on tactical vehicles to contain 155mm or 8-inch projectiles once they have been removed from their shipping pallets at the field artillery firing battery area during peacetime training exercises and in wartime. It is expected that throughout its life cycle the racks will be subjected to direct or indirect natural and induced environments.

6.1.2 120mm LRRS. The intent of the Loose Round Restraint System (LRRS), is to provide reusable dunnage on tactical vehicles to contain 120mm mortar rounds once they have been removed from shipping boxes but still contained within the fiberboard packing tubes. The tubes are placed vertically into the rack with the tube caps up. Once the cap is removed, the round is accessible directly from the tube or the tube removed from the rack. This action will take place at the heavy mortar firing battery area during peacetime training exercises and in wartime. It is expected that throughout its life cycle the racks will be subjected to direct or indirect natural and induced environments.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this and the item detail specifications.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

c. Quantity required and delivery schedules.

d. Serialization requirements, if applicable.

e. Quality Conformance Inspection, if other than specified in Section 4 of item detail specification.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:57Z
Check the source to verify that this is the current version before use.

f. Requirements for submission of first article sample.
g. Packaging requirements, if other than specified in Section 5 of item detail specification.

h. Certificates of conformance for each lot or shipment of product (see 4.1.2).

6.3 Data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Ref. Para.	DID Number	DID Title	Suggest Tailoring
3.7	DI-M-30413	Quality Program Plan	Delete USAF Bul 515

6.4 Definitions.

6.4.1 Loose projectile restraint system (LPRS). Refers to one or more 155mm projectile (9, 15, or 25 round) rack assemblies or 8-inch projectile 9 round rack assemblies (see Table VIII) restrained in a military tactical cargo vehicle by use of 1-3/4-inch nylon cargo tiedown straps. The projectiles are placed vertically in these racks with the base of the projectile resting on the floor of the vehicle. Tiedown procedures have been developed, tested and approved by the U.S. Army Defense Ammunition Center and School (USADACS).

6.4.2 Loose round restraint system (LRRS). Refers to one or more 120mm mortar 36 round rack assemblies (see Table VIII)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

restrained in a military tactical vehicle by use of 1-3/4 inch nylon cargo tiedown straps. The mortar rounds in their fiberboard packaging tubes are placed vertically in these racks with the base of the tube resting on the floor of the vehicle. Tiedown procedures have been developed, tested and approved by USADACS.

6.4.3 Divider. Refers to any of the specifically designed plastic blow-molded dividers used by both LPRS and LRRS which are used to make up the several divider racks. See Table VIII for physical data.

TABLE VIII. Blow molded plastic divider data

Restraint system	Dwg No.	Divider Length	Divider Height	Divider Width	Nominal wall thickness	Approx. weight lbs.	(Dividers) per rack assy combinations
155mm LPRS	AC200-000402	30-3/8	16	1-1/8	1/8	4.2	(8) 9 Rd Assy
							(6) 15 Rd Assy
							(0) 25 Rd Assy
155mm LPRS	AC200-000403	46-3/4	16	1-1/8	1/8	6.4	(0) 9 Rd Assy
							(4) 15 Rd Assy
							(12) 25 Rd Assy
8-inch LPRS	AC200-000438	40-3/4	22	1-1/2	1/8	8.8	(8) 9 Rd Assy
120mm LRRS	AC200-000591	47-5/8	20	1-1/8	1/8	8.6	(14) 36 Rd Assy

6.4.4 Plastic divider rack or rack. Refers to the several plastic divider rack assemblies listed in Table IX. See Table VIII for dividers per rack assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

TABLE IX. Divider rack assemblies

Restraint system	NSN	Illustrative drawing no.	Nominal size Length	Width	Height	Rounds per assy
155mm LPRS	None assigned	(TM9-2590 -210-10)	30-3/8	30-3/8	16	9
	2590-01 -223-2949	AC200000401	46-3/4	30-3/8	16	15
	None assigned	(TM9-2590 -210-10)	46-3/4	46-3/4	16	25
8-inch LPRS	2590-01 -295-5380	AC200000578	40-3/4	40-3/4	22	9
120mm LRRS	1090-01 -319-2625	AC200000597	47-5/8	47-5/8	20	36

6.4.5 First article regrind determination. The following discussion is to clarify that the fifth iteration of regrind will contain less than 2% material that has seen five or more melt cycles is based on an assumed 45% regrind rate. During the initial machine set-up, the parts and flashing produced will consist of 100% virgin material (designated as A0). Once regrind is generated from iteration A0 is introduced into the melt cycle, the parts and flashing will consists of 55% virgin and 45% "one time through" regrind (one time through is designated as A1). When the flashing from these parts is again introduced into the melt cycle, the resulting material composition of the parts and flashing consists of 55% virgin and 45% "once and twice through" regrind (twice through is A2, three times is A3, etc.). The first and subsequent iteration mixes are:

(1) .55A0 + .45A1

(2) .55A0 + .45(.55A1 + .45A2) = .55A0 + (.2475A1 + .2025A2)

(3) .55A0 + .45(.55A1 + .2475A2 + .2025A3) =
.55A0 + (.2475A1 + .1114A2 + .0911A3)

(4) .55A0 + .45(.55A1 + .2475A2 + .1114A3 + .0911A4) =
.55A0 + (.2475A1 + .1114A2 + .0501A3 + .010A4)

(5) .55A0 + .45(.55A1 + .2475A2 + .1114A3 + .0501A4 + .0410A5) =
.55A0 + (.2475A1 + .1114A2 + .0501A3 + .0225A4 + .0184A5)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

6.4.6 Fold Lines. Fold lines occur when the plastic is not uniformly distributed over the interior surface of the mold during forming, i.e., a fold occurs in the hot plastic bag (referred to in industry as the parison). Since there is no positive mechanical welding of the plastic along the fold, the fold line is of an unknown structural value. Additionally, the fold line robs the plastic available for distribution to the extremities of the mold. This can result in thin soft spots in the divider side walls to the extent of being translucent when sectioned and viewed from the inside.

6.5 Standard measuring and test equipment. Where standard measuring and test equipment (SMTE) is specified as the method of inspection, the contractor may use any type of industry-developed, commercially-available, multi-usage equipment or special inspection and test equipment approved by the government. The contractor shall be responsible for utilizing SMTE, which will assure full form for the entire surface of the specified characteristic.

6.6 Submission of contractor inspection designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAR-I. Picatinny Arsenal, NJ 07806-5000. This address will be specified on the contract Data Requirements List, DD Form 1423 in the contract.

6.7 Performance testing. Unless otherwise specified, the performance test will be conducted at the U.S. Army Defense Ammunition Center and School Test Facility located at Savanna, Illinois.

6.8 Tiedown procedures (see 4.5.1). The tiedown procedures and videotape(s) of previous testing can be obtained from Director, U.S. Army Defense Ammunition Center and School (USADACS), ATTN: SMCAC-DEV, Savanna, IL 61074-9639, or by phone to Mr. Jerry Krohn or Mr. John Sprague, (815) 273-8929.

6.9 Submission of alternative quality conformance provisions. Unless otherwise specified in the contract, proposed alternative quality conformances provisions will be submitted by the contractor for evaluation by the technical activity responsible for the preparation of this specification.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:57Z
Check the source to verify that this is the current version before use.

6.10 Subject term (key word) listing.

Process controls
Projectiles
Rounds
Vehicle Storage
Vehicle Transport

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian
Army-AR

Preparing Activity
Army-AR

(Project 2590-A021)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-R-48664B(AR)	930318

3. DOCUMENT TITLE: RESTRAINT SYSTEMS, AMMUNITION (USING BLOW MOLDED PLASTIC DIVIDERS)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6626	(2) AUTOVON DSN-880-6626
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.